

LARGE PARCEL-HUB TRACKING SIMULATION

Meridian Parcel Network Ltd Simulation

A tracking-first stress test of the THREADS temporal/relational reasoning engine across a fictional large UK parcel hub handling 500,000 parcels in one operating day.

Bottom line: THREADS loaded 2,506,891 unique tracking events and completed 141,280 graded checks with zero symbolic reasoning failures on correctly timestamped observed data. It preserved route-affecting ambiguity, resolved route-safe ambiguity, detected contradictions, reconstructed historical positions and recovered proof chains. Separate probes also exposed two production realities: missing scans cannot reveal unobserved physical movement, and stale records with incorrect effective timestamps can create false current state.

Prepared 12 September 2026. Meridian Parcel Network Ltd is fictional. The hub volume is calibrated against public UK parcel-hub capacities, not presented as data from a real carrier.

1. The simulated parcel company

Meridian Parcel Network Ltd is modelled as a national carrier operating a large central sorting hub. The hub processes **500,000 parcels across a 16-hour high-throughput operating window**, averaging about 31,250 parcels per hour. This is deliberately substantial but realistic: FedEx has announced planned UK hubs rated at 32,000 packs per hour; UPS says its East Midlands air hub can sort up to 22,500 parcels per hour; Royal Mail states that its super hubs average 73,700 parcels per hour and its largest automated sorting hub can process up to one million parcels per day.

500k

parcels tracked in the
simulated day

31.25k/h

average flow over the 16-
hour window

80

delivery routes represented

2.51m

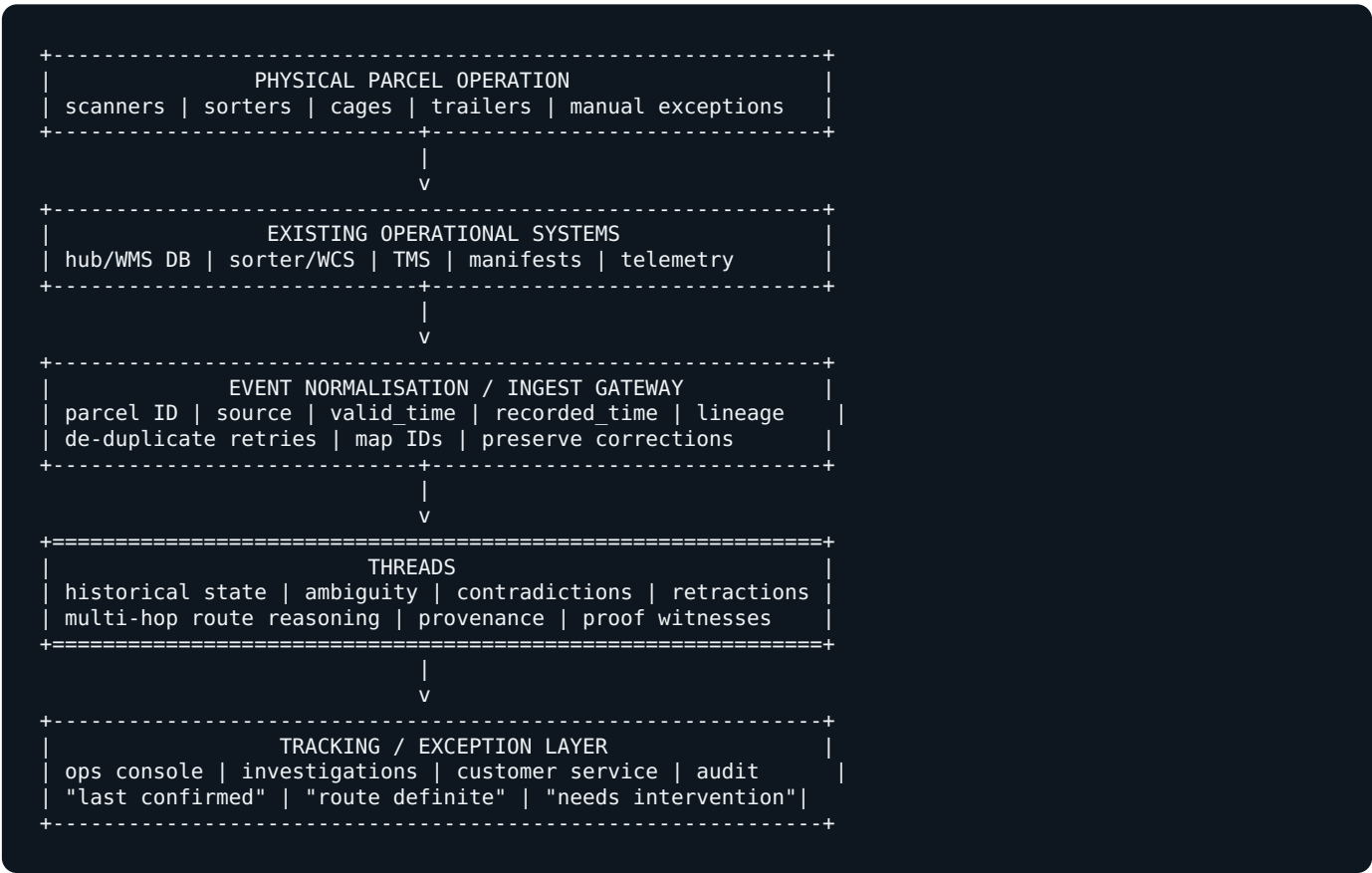
unique operational tracking
events

Physical process represented

```
graph TD
    A[INBOUND TRAILER] --> B[INBOUND DOOR]
    B --> C[INDUCTION SCAN]
    C --> D[AUTOMATED SORTER]
    D --> E[CHUTE]
    E --> F[DISPATCH CAGE]
    F --> G[OUTBOUND TRAILER]
    G --> H[DELIVERY ROUTE]
```

The simulation does **not** ask THREADS to control conveyors, generate labels, schedule drivers or optimise routes. It tests THREADS as the tracking-reconciliation layer that interprets the records those systems produce.

2. Where THREADS would sit



Placement principle: THREADS should sit beside the operational stack, not in the hard-real-time sorter-control loop. The existing systems create observations; THREADS determines what can be defensibly concluded from all observations together.

What each layer is responsible for

Layer	Job	Not THREADS's job
Scanners / sorters / telemetry	Observe physical movement	THREADS cannot observe a movement that generated no event.
WCS / WMS / TMS	Execute sortation, dispatch and transport	THREADS should not drive motors, belts or routing actuators.
Normalisation gateway	Preserve event meaning and timestamps	Do not convert message-arrival time into physical effective time.
THREADS	Reconcile history, alternatives, contradictions and evidence	It is not a replacement for the operational systems above.
Tracking / investigations	Present conclusions and exceptions to people or APIs	Should label inferred state accurately, e.g. "last confirmed".

3. What was made deliberately messy

The benchmark was designed to resemble imperfect tracking data rather than a clean sequence of one scan per parcel. The complete event stream was globally shuffled before ingestion so arrival order could not be relied upon.

Injected condition	Count	Why it matters operationally
Duplicate / retry messages	8,049	Network retries and repeated scanner submissions should not manufacture extra history.
Transient optical misreads	1,521	A bad sorter read is followed milliseconds later by a confirming scanner.
Targeted corrections / retractions	1,200	A supervisor retracts a specific wrong manual location without deleting valid independent evidence.
Dual-container ambiguity	600	Two plausible dispatch observations exist simultaneously.
Explicit contradictions	180	The same proposition is both positively and negatively observed.
Stale manifest records	250	Tests whether old data is harmless when its effective time is encoded correctly, and dangerous when it is not.
Completely missing final scans	300	The parcel physically moves, but no observation records the final move.

2,514,940

raw event messages generated

2,506,891

unique events stored after retry de-duplication

The event count is larger than the number of parcels because a parcel accumulates multiple observations and relations: inbound door, induction point, sort chute, cage, trailer and route, plus exceptional/corrective records.

4. Benchmark result

0 symbolic reasoning failures on correctly observed/ timestamped data	141,280 graded checks in the main run	11.97s load time for 2.51m unique events in this environment	1.54 GB peak resident memory during the run
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Capability tested	Result
Normal current-location tracking over sampled parcels	PASS
Historical reconstruction at multiple earlier stages	PASS
Duplicate event idempotency	PASS 8,049 retries did not become distinct history.
Same-route ambiguity	300 / 300 downstream route resolved definitively.
Different-route ambiguity	300 / 300 remained ambiguous rather than being guessed.
Explicit contradictions	180 / 180 returned contradictory state.
Three-hop route reasoning	PASS parcel -> cage -> trailer -> route.
Proof/evidence recovery	200 / 200 recovered a three-edge witness.

Throughput observed: 209,505 unique events/second during loading and about 174,783 sampled checks/ second during the query phase. These are benchmark measurements from the current Python implementation in this environment, not production service-level guarantees.

Important wording: the benchmark's 141,280 checks also include the separate missing-scan observability probe. Those 300 parcels intentionally have physical truth that is absent from the event stream. They are not counted as symbolic reasoning failures, because the engine was never given evidence of the move.

5. Why THREADS is useful for tracking

Case A - exact location uncertain, business route definite

```
Scanner A: parcel P123 -> cage CG80
Scanner B: parcel P123 -> cage CG160

CG80 -> trailer T40 -> route Manchester
CG160 -> trailer T120 -> route Manchester
```

A conventional exception screen sees two cage locations and may simply mark the parcel as discrepant. THREADS preserves the cage ambiguity but can still conclude that **both valid alternatives feed the same Manchester route**.

THREADS result

Exact cage: AMBIGUOUS
Route: ANSWER - Manchester

The operator can often continue without investigating the exact cage immediately.

Operational consequence

The system distinguishes an uncertainty that matters from one that does not matter to the current decision.

Case B - uncertainty changes the route

```
Scanner A: parcel P456 -> cage CG80 -> Manchester
Scanner B: parcel P456 -> cage CG81 -> Birmingham
```

THREADS result: route remains AMBIGUOUS. This is an exception worth intervention because the unresolved observation changes the business outcome.

The benchmark created 300 cases of each type. THREADS resolved all 300 same-route cases and preserved all 300 different-route cases as ambiguous.

6. Without THREADS vs with THREADS

Tracking problem	Without a reconciliation layer	With THREADS
Two scanners disagree	Show latest row, apply ad-hoc source priority, or open a manual exception.	Preserve both possibilities and determine whether they affect the downstream answer.
Later correction arrives	Overwrite the previous row or maintain custom audit tables.	Targeted retraction remains part of reproducible history.
Customer asks where parcel was at 05:17	Analyst reconstructs scan logs across systems.	Historical query is native to the temporal ledger.
Two sources directly conflict	Often represented as one current value plus an error note.	CONTRADICTED remains a first-class answer.
Investigator asks "why?"	Find and compare individual scan/manifests manually.	Proof/provenance can return the witness records behind a conclusion.
500,000-parcel day	Total data volume can make manual exception work expensive even if ordinary lookups are easy.	Each parcel query touches a small relevant history; benchmark query rate stayed high at 2.5m events.

The potential product difference: ordinary tracking systems are good at recording scans. THREADS is useful when the question becomes, "Given several imperfect scan histories, what can we actually establish, what remains uncertain, and which evidence supports it?"

Where time could be saved

- Fewer manual investigations for discrepancies that reconverge to the same route.
- Faster reconstruction of historical parcel state for claims, complaints and operational reviews.
- Less time collating evidence from WMS, scanner logs, manifests and trailer systems.
- Earlier prioritisation: route-affecting ambiguity can be escalated while route-safe ambiguity can be deferred.

7. Could this save money?

Potentially, yes - through exception handling and investigation labour, not through replacing the sorter. The benchmark does not contain real payroll or claim data, so the report does not claim a measured ROI. The calculation below is an illustrative way a carrier could evaluate a pilot.

100

illustrative discrepancy investigations
per day

10 min

illustrative time saved per investigation

16.7 h

staff time released per operating day

$100 \text{ investigations/day} \times 10 \text{ minutes saved} = 1,000 \text{ minutes/day}$
 $= 16.7 \text{ staff-hours/day}$

At an illustrative fully-loaded cost of GBP 25/hour:

$16.7 \times \text{GBP } 25 = \text{about GBP } 417/\text{day}$

$\text{GBP } 417 \times 260 \text{ days} = \text{about GBP } 108,000/\text{year}$

This is not a claim that THREADS saves GBP 108k/year. It is a pilot-sizing example. A real carrier would measure actual exception volume, minutes per investigation, staff cost, false-claim exposure, misroute recovery cost and customer-service contacts.

Other possible economic benefits

- **Avoided unnecessary interventions:** do not stop a parcel merely because exact cage identity is ambiguous if every valid cage feeds the same route.
- **Faster claims investigation:** reconstruct what was last confirmed and show the supporting scan chain.
- **Reduced false certainty:** do not silently turn conflicting observations into one apparently authoritative location.
- **Better root-cause analysis:** distinguish repeated scanner errors, stale manifests, manual corrections and missing observations.

8. Two limits a real deployment must respect

Limit 1 - missing scans are missing reality

For 300 deliberately selected parcels, the physical parcel moved from a sort chute into a dispatch cage, but the final scan message was removed from the event stream. THREADS then reported the previous chute for all 300.

This is not a reasoning bug. No event-driven tracking engine can know a physical movement that no sensor, person or system recorded. A production UI should therefore say "**last confirmed location**", include the age of the observation, and flag when an expected next scan is overdue.

```
Physical reality:      P499712 -> CG112
Last observed evidence: P499712 -> CH192

Recommended UI wording:
Last confirmed: CH192
Last observed: 7 minutes ago
Expected next event: dispatch-cage scan
Status: stale / investigate if threshold exceeded
```

Limit 2 - effective time must not be confused with arrival time

A separate probe replayed 250 old manifest locations but incorrectly timestamped them as if they became physically true when the messages arrived. All 250 superseded the genuine location.

250 / 250 false current locations in the bad-timestamp probe. The engine followed the supplied temporal semantics correctly; the integration supplied the wrong semantics.

A deployment should carry at least:

```
valid_time    = when the parcel was actually observed in that state
recorded_time = when the platform received or stored the message
```

This is the strongest production requirement exposed by the simulation.

9. What THREADS should and should not become

Good fit

- Cross-system parcel tracking reconciliation
- Historical incident reconstruction
- Misroute and exception investigation
- Claims evidence and provenance
- Contradictory scanner/manifest handling
- Container/trailer/route reasoning
- Tracking API that returns uncertainty explicitly

Not the target

- Conveyor PLC control
- Real-time motor/sorter actuation
- Route optimisation
- Driver scheduling
- Barcode recognition itself
- GPS hardware
- Label printing

Production engineering still required

Area	Requirement before deployment
Persistence	Durable storage, recovery, backup and replay rather than an in-memory-only process.
Temporal model	Add explicit valid/effective time and recorded/arrival time semantics.
Concurrency	Test simultaneous high-rate writers and readers under realistic service architecture.
Connectors	Adapters for scanner bus, WMS/WCS/TMS, manifests and telemetry with stable identity mapping.
Operational semantics	Expose "last confirmed" and staleness rather than claiming unobserved physical truth.
Resource guards	Keep a reasoning-cost/branch budget for pathological many-way unresolved uncertainty.
Observability	Metrics, alerting, audit logs, tracing and service health monitoring.

Recommendation: the tracking use case is credible enough for a pilot. The next useful milestone is not a larger synthetic parcel count; it is a production-shaped service prototype with durable storage, bitemporal ingestion and a real exception workflow.

10. Accuracy and completeness check

Check	Status
Benchmark rerun before report generation	DONE 500,000 parcels; 2,506,891 unique events; 141,280 checks.
Main reasoning failure count	VERIFIED 0 failures excluding the deliberately unobserved physical-move probe.
Same-route reconvergence	VERIFIED 300 / 300.
Different-route ambiguity	VERIFIED 300 / 300.
Contradiction detection	VERIFIED 180 / 180.
Proof recovery	VERIFIED 200 / 200.
Missing-scan observability probe	VERIFIED 300 / 300 physical final moves were unknowable from the supplied records.
Bad timestamp probe	VERIFIED 250 / 250 stale records produced false current location when encoded with wrong effective time.
Scale calibration	CHECKED Simulation throughput is within public large-hub ranges cited below.
Financial examples	ILLUSTRATIVE Explicitly labelled as pilot-sizing examples, not measured savings.

Benchmark provenance

The test used the current THREADS package source available in the working environment and the parcel benchmark file `parcel_hub_tracking_benchmark_500k_corrected.py`. The event stream was shuffled before insertion. Expected physical timelines and route assignments were generated independently inside the simulation and compared with THREADS results.

Sources used only to calibrate hub scale

FedEx, 9 May 2025: planned UK hubs each rated to sort 32,000 packs/hour.

<https://newsroom.fedex.com/newsroom/europe-english/fedex-commits-to-uk-growth-with-plans-unveiled-for-two-new-state-of-the-art-logistics-hubs>

UPS, 6 September 2021: East Midlands air hub can sort up to 22,500 parcels/hour.

<https://about.ups.com/gb/en/newsroom/press-releases/customer-first/uk-air-hub-expansion.html>

Royal Mail Group careers pages, accessed September 2026: super hubs average 73,700 parcels/hour; largest automated sorting hub cited at up to 1 million parcels/day.

<https://careers.royalmailgroup.com/parcelforce>

<https://careers.royalmailgroup.com/gb/en/engineering>

Meridian Parcel Network Ltd is fictional. This report evaluates a synthetic benchmark and should not be read as a claim of field deployment, carrier endorsement, production SLA or measured commercial ROI.